

Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r, \theta, t) = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left\{ -\frac{\omega}{c} \sin \left[\omega \left(t - \frac{r}{c} \right) \right] + \frac{1}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right] \right\}$$

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin \left[\omega \left(t - \frac{r}{c} \right) \right] \hat{z}$$

satisfy the Lorenz gauge condition.

Solution:

The Lorenz gauge condition is

$$\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Calculating the time derivative of V

$$\frac{\partial V}{\partial t} = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left\{ -\frac{\omega}{c} \sin \left[\omega \left(t - \frac{r}{c} \right) \right] + \frac{1}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right] \right\}$$

Calculating the divergence of $\mathbf{A}$

$$\nabla \cdot \mathbf{A} =$$

□

Problem 11.2

Problem 11.3

Problem 11.4

Problem 11.5

Problem 11.6

Problem 11.7

Problem 11.8

Problem 11.9

Problem 11.10

Problem 11.11

Problem 11.12

Problem 11.13

Problem 11.14

Problem 11.15

Problem 11.16

Problem 11.17

Problem 11.18

Problem 11.19

Problem 11.20

Problem 11.21

Problem 11.22

Problem 11.23

Problem 11.24

Problem 11.25

Problem 11.26

Problem 11.27

Problem 11.28

Problem 11.29

Problem 11.30

Problem 11.31

Problem 11.32

Problem 11.33

Problem 11.34

Problem 11.35

Problem 11.36